

S2D-06: ArrayMovingSum for Extended Timeframes

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Overview

Davis Anomaly Detectors have a maximum sliding window of 60 minutes. For alerts that need to evaluate longer timeframes while still using continuous monitoring, the `arrayMovingSum` function provides a solution.

This notebook explains how to use `arrayMovingSum` to create rolling aggregations that can be used with Davis Anomaly Detectors.

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Prerequisites

Requirement	Details
Dynatrace Environment	SaaS with Grail enabled
Permissions	<code>logs.read</code>
Knowledge	Basic DQL and array functions

Learning Objectives

By the end of this notebook, you will be able to:

1. Understand the `arrayMovingSum` function
2. Create rolling sum queries for extended timeframes
3. Apply this pattern to Davis Anomaly Detectors
4. Recognize limitations and alternatives

The Problem

Many Splunk alerts evaluate over timeframes longer than 1 hour:

- "Alert if more than 500 errors in the last 4 hours"
- "Alert if average response time exceeds 2s over the past 2 hours"

Davis Anomaly Detectors have a **maximum sliding window of 60 minutes**.

The Solution: ArrayMovingSum

The `arrayMovingSum` function calculates a rolling sum across timeseries data points, allowing you to aggregate values over longer periods while maintaining 1-minute granularity.

How ArrayMovingSum Works

```
arrayMovingSum(array, window)
```

Parameter	Description
array	The timeseries array to aggregate
window	Number of data points to sum

Timeframe Calculation

The effective timeframe depends on:

- **Interval:** The timeseries interval (e.g., 1m)
- **Window:** The number of points to sum

```
Effective Timeframe = Interval × Window
```

Example: `interval:1m` with `window:60` = 60-minute rolling sum

Basic Example

Create a 60-minute rolling sum of error logs:

```
// 60-minute rolling sum of error logs
fetch logs, from:-24h
| filter loglevel == "ERROR"
| makeTimeseries error_count = count(), interval:1m
| fieldsAdd error_count_last_1h = arrayMovingSum(error_count, 60)
| fieldsRemove error_count
```

Understanding the Output

Each data point in `error_count_last_1h` represents the **sum of errors in the preceding 60 minutes**.

Time	error_count (per minute)	error_count_last_1h (rolling 60m)
10:00	5	120 (sum of 9:01-10:00)
10:01	3	118 (sum of 9:02-10:01)
10:02	8	123 (sum of 9:03-10:02)

Limitations

Maximum Window Size: 60

The `arrayMovingSum` window parameter has a **maximum value of 60**.

With a 1-minute interval, this means:

- **Maximum effective timeframe: 60 minutes**

Increasing the Interval?

You could increase the interval to extend the timeframe:

Interval	Window	Effective Timeframe
1m	60	60 minutes
5m	60	300 minutes (5 hours)
10m	60	600 minutes (10 hours)

However: Davis Anomaly Detectors require a **1-minute interval**, so this workaround only applies to dashboards and reports, not continuous alerting.

Using with Davis Anomaly Detectors

For Davis Anomaly Detectors, you must use `interval:1m` and `window:60` :

```
// Query for Davis Anomaly Detector with 60-minute rolling sum
fetch logs, from:-24h
| filter loglevel == "ERROR"
| filter matchesPhrase(k8s.deployment.name, "checkout-service")
| makeTimeseries error_count = count(), by:{dt.entity.cloud_application},
interval:1m
| fieldsAdd error_count_1h = arrayMovingSum(error_count, 60)
| fieldsRemove error_count
```

Configuring the Detector

When creating the Davis Anomaly Detector:

Setting	Value	Rationale
Threshold	Total errors in 60 minutes	e.g., 100
Sliding Window	1 minute	Each point already contains 60m of data
Violating Samples	1	Rolling sum already aggregates

Key Insight: Since each data point already represents 60 minutes of data, you typically want minimal sliding window and violating samples.

Dashboard/Report Example (Longer Timeframes)

For dashboards and reports (not Davis Anomaly Detectors), you can use longer intervals:

```
// 4-hour rolling sum for dashboard visualization
// Uses 4-minute interval x 60 window = 240 minutes (4 hours)
fetch logs, from:now()-24h
| filter loglevel == "ERROR"
| makeTimeseries error_count = count(), interval:4m
| fieldsAdd error_count_4h = arrayMovingSum(error_count, 60)
| fieldsRemove error_count
```

Other Array Functions

Related functions for rolling calculations:

Function	Description
<code>arrayMovingSum</code>	Rolling sum
<code>arrayMovingAvg</code>	Rolling average
<code>arrayMovingMin</code>	Rolling minimum
<code>arrayMovingMax</code>	Rolling maximum
<code>arrayMovingMedian</code>	Rolling median

```
// Rolling average response time over 30 minutes
fetch logs, from:-24h
| filter isNotNull(response_time)
| makeTimeseries avg_response = avg(response_time), interval:1m
| fieldsAdd avg_response_30m = arrayMovingAvg(avg_response, 30)
```

When ArrayMovingSum Isn't Enough

If you need timeframes exceeding 60 minutes with Davis Anomaly Detectors, you have these options:

Approach	Use Case	Trade-offs
Workflow-based alerting	Any timeframe	No Davis AI, consumes workflow hours
Metric extraction	High-volume queries	Requires OpenPipeline config
Reduce threshold	Proportional reduction	May increase false positives

Proportional Threshold Reduction Example

If Splunk alert is: 500 errors in 4 hours

Convert to 1-hour equivalent:

- $500 \div 4 = 125$ errors per hour
- Set Davis threshold to 125 with arrayMovingSum over 60 minutes

Caveat: This assumes errors are evenly distributed, which may not match actual patterns.

Summary

Scenario	Solution
Alert over ≤ 60 minutes	Standard Davis Anomaly Detector
Alert over exactly 60 minutes	ArrayMovingSum with window:60
Alert over > 60 minutes	Use Workflows (see S2D-05)
Dashboard visualization > 60 minutes	ArrayMovingSum with larger interval

Next Steps

- **S2D-07: Metric Creation** - For performance-critical alerting queries
- **S2D-05: Workflow Alerts** - For timeframes exceeding 60 minutes

References

- [ArrayMovingSum Documentation](#)
- [DQL Array Functions](#)
- [Davis Anomaly Detectors](#)

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